

Mathematics Stage 3 -- Year B -- Unit 36 Fractions & Chance

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency through exploring, applying and communicating mathematical reasoning.
MA3-RQF-01	Compares and orders fractions with denominators of 2, 3, 4, 5, 6, 8 and 10.
MA3-RQF-02	Determines fractions of measures and quantities; adds and subtracts fractions.
MA3-RN-03	Determines percentages of quantities, and finds equivalent fractions and decimals for benchmark percentage values.
MA3-CHAN-01	Conducts chance experiments and quantifies the probability using fractions.
MA4-FRC / PRO	Stage 4 (group A): operates with fractions of different denominators and multiplies fractions (FRC); finds probabilities of complementary events (PRO).

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Fractions & benchmarks (recall & single procedures) Recognise the number 1 as the whole; compare halves/quarters of different-sized wholes (RQF-A). Recall benchmark fraction-decimal-percentage equivalence ($\frac{1}{2} = 50\% = 0.5$, $\frac{1}{4} = 25\%$, $\frac{3}{4} = 75\%$). Find a half, quarter, fifth or tenth of a whole number. Subtract a unit fraction from a whole number using the complement principle (RQF-A).	Fractions & chance (applying) Build the whole quantity from a non-unit fractional part. Compare fractions with related denominators (2,4,8; 3,6; 5,10) using area and discrete models. Order common fractions with related denominators using diagrams and number lines. Use equivalence to add and subtract fractions with related denominators. Solve word problems involving adding or subtracting fractions with related denominators. Assign probabilities using fractions and benchmark fractions, decimals and percentages.	Extended -- Stage 4 Fractions & Probability Add and subtract fractions with different denominators by finding a common denominator (MA4-FRC-C-01). Multiply a fraction by a whole number and by another fraction (MA4-FRC-C-01). List the sample space and find probabilities of complementary events, $P(\text{not } A) = 1 - P(A)$ (MA4-PRO-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students describe fractions, compare them using models, and explain how likely a chance outcome is using fractions, decimals and percentages.
Understanding and fluency	Students find fractions of quantities, compare and order related denominators, and add and subtract fractions using equivalence.
Reasoning	Students justify the need for a common whole, use the complement principle, and reason about probabilities.
Problem solving	Students solve fraction word problems and chance problems, and (Stage 4) add/multiply unlike fractions and find complementary probabilities.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Studen ts who need further consol idation	
Next steps	

Resources

See Appendix for Resources
Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 36 | NESA Mathematics K-10 Syllabus (2022)

Unit 36 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Recognise the whole; compare halves and quarters of different-sized wholes.	RQF-A: recognise the number 1 as the whole; compare halves/quarters of different-sized wholes MA3-RQF-02 Year A	Mini Lesson 1		
R2	Recall benchmark fraction-decimal-percentage equivalence.	RN-B: % means percent and 100% is the whole; recall equivalent percentages, decimals and fractions MA3-RN-03 Year B	Mini Lesson 3		
R3	Find a half, quarter, fifth or tenth of a whole number.	RQF / RN-B: find fractional quantities of whole numbers (halves, quarters, fifths, tenths); find 10% of a quantity MA3-RQF-02 / MA3-RN-03 Year B	Mini Lesson 4		
R4	Subtract a unit fraction from a whole number (complement principle).	RQF-A: subtract a unit fraction from any whole number including 1 (complement principle) MA3-RQF-02 Year A	Mini Lesson 7		
Y1	Build the whole quantity from a non-unit fractional part.	RQF-B: build up to the whole from a given fractional part; generate the whole from non-unit fractional parts MA3-RQF-02 Year B	Mini Lesson 2		
Y2	Compare fractions with related denominators using models.	RQF-B: compare/represent fractions with denominators 2,4,8; 3,6; 5,10 (area & discrete models); subdivide a rectangle to show the relationship MA3-RQF-01 Year B	Mini Lesson 5		
Y3	Order common fractions with related denominators.	RQF-B: order common fractions with related denominators using diagrams and number lines MA3-RQF-01 Year B	Mini Lesson 6		
Y4	Add and subtract fractions with related denominators using equivalence.	RQF-B: use equivalence to add and subtract fractional quantities with related denominators MA3-RQF-02 Year B	Mini Lesson 8		
Y5	Solve word problems adding or subtracting fractions with related denominators.	RQF-B: solve word problems involving adding/subtracting fractional quantities with related denominators MA3-RQF-02 Year B	Mini Lesson 9		
Y6	Assign probabilities using fractions and benchmark FDP.	CHAN-B: assign probabilities to outcomes using fractions; use benchmark fractions/decimals/percentages for likelihood MA3-CHAN-01 Year B	Mini Lesson 10		
G1	Add and subtract fractions with different denominators.	Stage 4 FRC-A: compare/order fractions with different denominators; add/subtract by finding a common denominator MA4-FRC-C-01 Stage 4 (group A)	Mini Lesson 11		
G2	Multiply a fraction by a whole number and by another fraction.	Stage 4 FRC-A: multiply fractions; apply knowledge of fractions of quantities to solve problems MA4-FRC-C-01 Stage 4 (group A)	Mini Lesson 12		

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
G3	List the sample space and find probabilities of complementary events.	Stage 4 PRO-A: list the sample space; determine probabilities for complementary events, $P(\text{not } A) = 1 - P(A)$ MA4-PRO-C-01 Stage 4 (group A)	Mini Lesson 13		

Syllabus outcomes and content from Mathematics K-10 Syllabus (2022) copyright NSW Education Standards Authority (NESA) for and on behalf of the Crown in right of the State of New South Wales, 2024. RQF = Representing quantity fractions. RN = Represents numbers (percentages). CHAN = Chance. FRC = Fractions, decimals & percentages (Stage 4). PRO = Probability (Stage 4). A/B = syllabus content groups (this unit genuinely mixes A and B fraction content).

Unit 36 -- Mini Lesson Sequence

Fractions & Chance | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-7

Yellow band -- On-level -- Mini Lessons 2-10

Green band -- Extension -- Mini Lessons 11-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 36. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 The whole and different-sized wholes **[HANDS-ON]**

R1 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min | **Outcomes: R1** | Syllabus: MA3-RQF-02, MAO-WM-01 | RQF-A: the number 1 as the whole; compare halves/quarters of different wholes

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Compare fractions with different-sized wholes | Key resources: Resource 1 - build it up, paper strips, fraction tiles

Physical materials needed: paper strips, fraction tiles

Learning intention

Students are learning to:
recognise the number 1 as the whole and compare halves and quarters of different-sized wholes

Success criteria

Students can:

- * I know the number 1 represents the whole.
- * I can see that half of a big shape is bigger than half of a small shape.
- * I can explain why fractions need a common whole to compare fairly.

Mini masterclass structure (25-30 min)

Activate (4 min): Show half of a large pizza and half of a small pizza. Ask: are these halves equal? Why not?

Model (10 min): Fold paper strips of different lengths into halves and quarters. Compare the actual sizes; establish that fractions only compare fairly when the whole (1) is the same.

Guided practice (10 min): Students compare halves and quarters of different-sized wholes and justify which is larger.

Check (4 min): Exit question: is $\frac{1}{2}$ of a 10 cm strip the same length as $\frac{1}{2}$ of a 20 cm strip? Explain.

Key vocabulary

whole, one, fraction, half, quarter, compare, equal, common whole

Differentiation

Support: Use same-size strips first to see equal halves, then introduce different-size wholes.
Extension: Compare thirds and sixths of different-sized wholes and explain using the common whole.

Assessment

Teaching notes (annotate here)

Students compare fractions of different-sized wholes — links to R1 in the SOLO Show task.

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Building the whole from a part [HANDS-ON]

Y1 | Yellow band | DoE Lesson 1 (adapted)

Duration: 25-30 min

Outcomes: Y1

Syllabus: MA3-RQF-02, MAO-WM-01 | RQF-B: generate the whole from a non-unit fractional part

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Build up to the whole | Key resources: Fraction strips, number lines, counters

Physical materials needed: fraction strips, number lines, counters

Learning intention

Students are learning to:
generate the whole quantity from a given non-unit fractional part

Success criteria

Students can:

- * I can find one part when I know a non-unit fraction (e.g. if $\frac{3}{4}$ is 18, one quarter is 6).
- * I can build up to the whole from the part.
- * I can check by finding the fraction of my answer.

Mini masterclass structure (25-30 min)

Activate (4 min): Tell students $\frac{3}{4}$ of a jar of marbles is 18. Ask how many marbles are in the whole jar.

Model (10 min): Use a fraction bar: $\frac{3}{4} = 18$, so one quarter = $18 \div 3 = 6$, and the whole = $6 \times 4 = 24$. Repeat with $\frac{2}{5} = 10 \rightarrow 25$.

Guided practice (10 min): Students find the whole from a non-unit fractional part using a bar model, then check.

Check (4 min): Exit question: $\frac{2}{3}$ of a length is 12 m. How long is the whole length?

Key vocabulary

fraction, non-unit fraction, part, whole, unit fraction, build up, divide, multiply

Differentiation

Support: Draw the fraction bar and find one part first; use counters for the part.
Extension: Find the whole from parts like $\frac{3}{5}$ and $\frac{5}{8}$, then find a different fraction of that whole.

Assessment

Students generate the whole from a fractional part — links to Y1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Fraction, decimal and percentage benchmarks

R2 | Red band | DoE Lesson 2 (adapted)

Duration: 25 min | **Outcomes:** R2 | Syllabus: MA3-RN-03, MAO-WM-01 | RN-B: % means percent; recall equivalent FDP benchmarks

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Benchmark fractions, decimals and percentages | Key resources: Hundred grids, fraction-decimal-percentage cards

Learning intention		Success criteria
Students are learning to: make connections between benchmark fractions, decimals and percentages		Students can: * I know % means per cent and 100% is the whole. * I can recall $\frac{1}{2} = 50\% = 0.5$, $\frac{1}{4} = 25\% = 0.25$ and $\frac{3}{4} = 75\% = 0.75$. * I can match a fraction, a decimal and a percentage that are equal.
Mini masterclass structure (25 min)		
Activate (3 min): Shade $\frac{1}{2}$ of a hundred grid. Ask what percentage and decimal that is. Model (10 min): Build a benchmark table: $\frac{1}{2} = 50\% = 0.5$; $\frac{1}{4} = 25\% = 0.25$; $\frac{3}{4} = 75\% = 0.75$; $\frac{1}{10} = 10\% = 0.1$. Show each on the grid. Guided practice (8 min): Students match fraction, decimal and percentage cards and complete a benchmark table. Check (4 min): Exit question: write $\frac{3}{4}$ as a decimal and a percentage.		
Key vocabulary		Differentiation
per cent, %, fraction, decimal, percentage, equivalent, benchmark, whole		Support: Use a shaded hundred grid so each percentage is visible; start with $\frac{1}{2}$ and $\frac{1}{4}$. Extension: Add $\frac{1}{5} = 20\%$ and $\frac{1}{3} \approx 33\%$ to the table and discuss the non-exact ones.
Assessment		Teaching notes (annotate here)
Students recall benchmark FDP equivalence — links to R2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Fractions of whole numbers [HANDS-ON]

R3 | Red band | DoE Lessons 3 & 5 (adapted)

Duration: 25-30 min

Outcomes: R3

Syllabus: MA3-RQF-02, MA3-RN-03, MAO-WM-01 | find halves, quarters, fifths and tenths of numbers

Linked to: DoE Lessons 3 & 5 (adapted) -- Lessons 3 & 5 -- Understanding tenths / tape diagrams | Key resources: Resource 14 - tape diagram, Resource 15 - lolly shop cards

Physical materials needed: tape diagram, lolly shop cards

Learning intention

Students are learning to:
find fractional quantities of whole numbers (halves, quarters, fifths and tenths)

Success criteria

Students can:

- * I can find a half or a quarter of a number by dividing.
- * I can find a fifth and a tenth of a number.
- * I know 10% of a number is the same as one-tenth of it.

Mini masterclass structure (25-30 min)

Activate (4 min): Use a tape diagram for 20 lolly cards. Ask for half, then a quarter.

Model (10 min): $\frac{1}{2}$ of 20 = 10 ($\div 2$); $\frac{1}{4}$ of 20 = 5 ($\div 4$); $\frac{1}{5}$ of 20 = 4 ($\div 5$); $\frac{1}{10}$ of 20 = 2 ($\div 10$). Connect $\frac{1}{10}$ to 10%.

Guided practice (10 min): Students find halves, quarters, fifths and tenths of whole numbers using tape diagrams.

Check (4 min): Exit question: find $\frac{1}{4}$ of 40 and $\frac{1}{10}$ of 50.

Key vocabulary

fraction, half, quarter, fifth, tenth, of, divide, tape diagram, 10%

Differentiation

Support: Use repeated halving (half, then half again for a quarter); use the tape diagram.
Extension: Find eighths from quarters and tenths from fifths; find 25% and 10% of money amounts.

Assessment

Students find fractions of whole numbers — links to R3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Comparing fractions (related denominators) [HANDS-ON]

Y2 | Yellow band | DoE Lesson 4 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-RQF-01, MAO-WM-01 | RQF-B: compare fractions with related denominators (area model)

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Compare common fractions with related denominators | Key resources: Resource 12 - area model fractions, Resource 13 - fraction wall

Physical materials needed: area model cards, fraction wall

Learning intention	Success criteria
Students are learning to: compare fractions with related denominators using an area model and a collection of objects	Students can: * I can use an area model or fraction wall to compare fractions. * I can compare fractions with related denominators (e.g. $\frac{1}{2}$ and $\frac{3}{8}$; $\frac{2}{3}$ and $\frac{5}{6}$). * I can rename a fraction to a related denominator to compare (e.g. $\frac{1}{2} = \frac{2}{4}$).

Mini masterclass structure (30 min)

Activate (4 min): Show $\frac{1}{2}$ and $\frac{3}{8}$ on a fraction wall. Ask which is bigger and how you can tell.

Model (12 min): Use an area model: rename $\frac{1}{2}$ as $\frac{4}{8}$, then compare with $\frac{3}{8}$. Subdivide a rectangle by length and width to see eighths from halves and quarters.

Guided practice (10 min): Students compare fractions with related denominators using the area model and fraction wall.

Check (4 min): Exit question: which is bigger, $\frac{2}{3}$ or $\frac{5}{6}$? Use a model to show why.

Key vocabulary	Differentiation
compare, fraction, related denominator, area model, fraction wall, equivalent, rename	Support: Provide a fraction wall; line up the two fractions and read off which is longer. Extension: Compare three related fractions and order them using the area model.
Assessment	Teaching notes (annotate here)
Students compare fractions with related denominators — links to Y2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Ordering fractions (related denominators) [HANDS-ON]

Y3 | Yellow band | DoE Lesson 5 (adapted)

Duration: 25-30 min **Outcomes:** Y3 Syllabus: MA3-RQF-01, MAO-WM-01 | RQF-B: order fractions with related denominators using number lines

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Ordering fractions with related denominators | Key resources: Number lines, Resource 16 - fraction cards, fraction wall

Physical materials needed: number lines, fraction cards

Learning intention

Students are learning to:
order common fractions with related denominators using diagrams and number lines

Success criteria

Students can:

- * I can place fractions with related denominators on a number line.
- * I can rename fractions to a common denominator to order them.
- * I can order a set of fractions from smallest to largest.

Mini masterclass structure (25-30 min)

Activate (4 min): Place $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{3}{4}$ on a 0-1 number line. Ask which is smallest.

Model (10 min): Order $\frac{2}{5}$, $\frac{1}{2}$ and $\frac{3}{5}$ by renaming to tenths ($\frac{4}{10}$, $\frac{5}{10}$, $\frac{6}{10}$) and placing on a number line.

Guided practice (10 min): Students order sets of fractions with related denominators using a number line and a fraction wall.

Check (4 min): Exit question: order $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{3}{8}$ from smallest to largest.

Key vocabulary

order, fraction, number line, related denominator, smallest, largest, rename, equivalent

Differentiation

Support: Provide a marked number line; rename to a common denominator first.
Extension: Order a mixed set including fractions greater than 1 (e.g. $\frac{5}{4}$).

Assessment

Students order fractions with related denominators — links to Y3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Subtracting a fraction from a whole [HANDS-ON]

R4 | Red band | DoE Lesson 6 (adapted)

Duration: 25 min | **Outcomes:** R4 | Syllabus: MA3-RQF-02, MAO-WM-01 | RQF-A: subtract a unit fraction from a whole (complement principle)

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Subtracting a unit fraction from a whole | Key resources: Resource 13 - fraction wall, number lines, Resource 18 - fractions

Physical materials needed: fraction wall, number lines

Learning intention	Success criteria
Students are learning to: subtract a unit fraction from a whole number using the complement principle	Students can: * I know that $1 = \text{a fraction} + \text{its complement}$ (e.g. $1 = \frac{1}{3} + \frac{2}{3}$). * I can subtract a unit fraction from 1 (e.g. $1 - \frac{1}{4} = \frac{3}{4}$). * I can subtract a unit fraction from a whole number (e.g. $3 - \frac{1}{4} = 2\frac{3}{4}$).

Mini masterclass structure (25 min)

Activate (4 min): Show a whole circle. Take away $\frac{1}{3}$. Ask how much is left and how you know.

Model (10 min): Use the complement: $1 - \frac{1}{3} = \frac{2}{3}$; $1 - \frac{1}{4} = \frac{3}{4}$. Then $3 - \frac{1}{4}$: keep 2 wholes, $1 - \frac{1}{4} = \frac{3}{4}$, so $2\frac{3}{4}$.

Guided practice (8 min): Students subtract unit fractions from 1 and from whole numbers using a fraction wall or number line.

Check (3 min): Exit question: $2 - \frac{1}{5} = ?$

Key vocabulary	Differentiation
whole, unit fraction, complement, subtract, left, number line, fraction wall	Support: Use a fraction wall to see the complement; start with subtracting from 1. Extension: Subtract non-unit fractions from whole numbers (e.g. $3 - \frac{2}{3}$).
Assessment	Teaching notes (annotate here)
Students subtract a unit fraction from a whole — links to R4 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Adding and subtracting fractions

Y4 | Yellow band | DoE Lessons 6 & 7 (adapted)

Duration: 30 min | **Outcomes:** Y4 | Syllabus: MA3-RQF-02, MAO-WM-01 | RQF-B: use equivalence to add/subtract related denominators

Linked to: DoE Lessons 6 & 7 (adapted) -- Lessons 6 & 7 -- Adding and subtracting fractions | Key resources: Fraction wall, Resource 18 - fractions

Learning intention		Success criteria	
Students are learning to: use equivalence to add and subtract fractions with related denominators		Students can: * I can add and subtract fractions with the same denominator. * I can rename a fraction to a related denominator before adding (e.g. $\frac{1}{2} = \frac{2}{4}$). * I can add fractions like $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$.	
Mini masterclass structure (30 min)			
Activate (4 min): Show $\frac{3}{8} + \frac{2}{8}$ on a fraction wall. Ask for the total. Model (10 min): Add same denominators: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$. Related: $\frac{1}{2} + \frac{1}{4} \rightarrow$ rename $\frac{1}{2}$ as $\frac{2}{4} \rightarrow \frac{2}{4} + \frac{1}{4} = \frac{3}{4}$. Subtract $\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$. Guided practice (12 min): Students add and subtract fractions with related denominators, renaming to a common denominator. Check (4 min): Exit question: $\frac{2}{3} + \frac{1}{6} = ?$ (rename $\frac{2}{3}$ as $\frac{4}{6}$).			
Key vocabulary		Differentiation	
add, subtract, fraction, denominator, equivalent, rename, common denominator, simplify		Support: Use a fraction wall; rename one fraction so both denominators match. Extension: Add three related fractions and simplify the answer.	
Assessment		Teaching notes (annotate here)	
Students add and subtract fractions with related denominators — links to Y4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 9 Fraction word problems

Y5 | Yellow band | DoE Lesson 7 (adapted)

Duration: 30 min | **Outcomes:** Y5 | Syllabus: MA3-RQF-02, MAO-WM-01 | RQF-B: word problems adding/subtracting related denominators

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Solving fraction word problems | Key resources: Bar models, Resource 18 - fractions

Learning intention		Success criteria
Students are learning to: solve word problems involving adding or subtracting fractions with related denominators		Students can: * I can draw a bar model to represent a fraction word problem. * I can decide whether to add or subtract the fractions. * I can solve the problem and check my answer.
Mini masterclass structure (30 min)		
Activate (4 min): Francis ate $\frac{3}{8}$ of a chocolate bar, then $\frac{1}{4}$ more. Ask how much he ate altogether. Model (10 min): Draw a bar model; rename $\frac{1}{4}$ as $\frac{2}{8}$; $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$. Then a 'how much is left' problem using subtraction. Guided practice (12 min): Students solve add/subtract fraction word problems with related denominators using bar models. Check (4 min): Exit question: a jug is $\frac{2}{3}$ full, then $\frac{1}{6}$ is poured out. How much is left?		
Key vocabulary		Differentiation
word problem, fraction, bar model, add, subtract, related denominator, total, left		Support: Provide a part-drawn bar model; help rename to the same denominator. Extension: Solve two-step fraction word problems and create one for a partner.
Assessment		Teaching notes (annotate here)
Students solve fraction word problems — links to Y5 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Chance with fractions [HANDS-ON]

Y6 | Yellow band | DoE Lesson 8 (adapted)

Duration: 30-35 min **Outcomes:** Y6 Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: assign probabilities using fractions and benchmark FDP

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Chance using fractions, decimals and percentages | Key resources: Spinners, coloured counters, bags, Resource - spinner bar model

Physical materials needed: spinners, coloured counters, bags

Learning intention

Students are learning to:
assign probabilities to outcomes using fractions and use benchmark FDP to describe likelihood

Success criteria

Students can:

- * I can write the probability of an outcome as a fraction.
- * I can assign probabilities when outcomes are not equally likely.
- * I can use benchmark fractions, decimals and percentages to describe how likely an outcome is.

Mini masterclass structure (30-35 min)

Activate (4 min): Show a spinner split into 4 equal parts (2 red, 1 blue, 1 green). Ask the chance of landing on red.

Model (12 min): $P(\text{red}) = \frac{2}{4} = \frac{1}{2} = 50\%$; $P(\text{blue}) = \frac{1}{4} = 25\%$. For an unequal spinner, use a bar model to read off the fractions.

Guided practice (14 min): Students assign probabilities to spinner and bag outcomes as fractions and describe each with a benchmark FDP.

Check (4 min): Exit question: a bag has 3 red and 1 blue counter. Write $P(\text{red})$ as a fraction and a percentage.

Key vocabulary

chance, probability, outcome, likely, fraction, equally likely, spinner, benchmark

Differentiation

Support: Use equal-section spinners first; count the favourable sections over the total.
Extension: Design a spinner to match given probabilities (e.g. $P(\text{red}) = \frac{3}{8}$).

Assessment

Students assign probabilities using fractions — links to Y6 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Fractions with different denominators (Stage 4)

G1 | Green band | Original mini lesson

Duration: 30-35 min

Outcomes: G1

Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4 FRC-A: add/subtract fractions with different denominators

Learning intention

Students are learning to:
add and subtract fractions with different (unrelated) denominators by finding a common denominator

Success criteria

Students can:

- * I can find a common denominator for two unlike fractions.
- * I can rename both fractions and then add or subtract.
- * I can simplify the answer where possible.

Mini masterclass structure (30-35 min)

Activate (4 min): Show $\frac{1}{3} + \frac{1}{4}$. Ask why we can't just add the tops and bottoms.

Model (12 min): Common denominator of 3 and 4 is 12: $\frac{1}{3} = \frac{4}{12}$, $\frac{1}{4} = \frac{3}{12}$, so $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$. Subtract $\frac{1}{2} - \frac{1}{3} = \frac{3}{6} - \frac{2}{6} = \frac{1}{6}$.

Guided practice (12 min): Students add and subtract fractions with different denominators by finding a common denominator.

Check (5 min): Exit question: $\frac{3}{4} - \frac{1}{3} = ?$

Key vocabulary

fraction, denominator, common denominator, equivalent, rename, add, subtract, simplify

Differentiation

Support: Provide a list of multiples to find the common denominator; use a fraction wall.
Extension: Add three unlike fractions and simplify; add mixed numbers.

Assessment

Students add/subtract unlike fractions — links to G1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Multiplying fractions (Stage 4)

G2 | Green band | Original mini lesson

Duration: 30 min | **Outcomes:** G2 | Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4 FRC-A: multiply fractions

Learning intention		Success criteria
Students are learning to: multiply a fraction by a whole number and by another fraction		Students can: * I can multiply a fraction by a whole number (e.g. $\frac{2}{3} \times 6 = 4$). * I can multiply two fractions by multiplying tops and bottoms. * I can connect 'of' to multiplication ($\frac{1}{2}$ of $\frac{3}{4} = \frac{1}{2} \times \frac{3}{4}$).
Mini masterclass structure (30 min)		
Activate (4 min): Ask what $\frac{2}{3}$ of 6 is, connecting 'of' to multiplication. Model (12 min): $\frac{2}{3} \times 6 = (2 \times 6) \div 3 = 4$. Multiply two fractions: $\frac{1}{2} \times \frac{3}{4} = (1 \times 3) / (2 \times 4) = \frac{3}{8}$. Guided practice (10 min): Students multiply fractions by whole numbers and by other fractions, simplifying where possible. Check (4 min): Exit question: find $\frac{3}{4} \times 8$ and $\frac{1}{3} \times \frac{1}{2}$.		
Key vocabulary		Differentiation
multiply, fraction, whole number, numerator, denominator, of, simplify, product		Support: Model 'of' with an area diagram; multiply tops and bottoms. Extension: Multiply a mixed number by a fraction and simplify.
Assessment		Teaching notes (annotate here)
Students multiply fractions — links to G2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 13 Probability and complementary events (Stage 4)

G3 | Green band | Original mini lesson

Duration: 30 min | **Outcomes: G3** | Syllabus: MA4-PRO-C-01, MAO-WM-01 | Stage 4 PRO-A: sample space; complementary events

Learning intention		Success criteria	
Students are learning to: list the sample space and find the probability of complementary events		Students can: * I can list all the possible outcomes (the sample space). * I know complementary events together cover everything (total = 1). * I can find $P(\text{not } A) = 1 - P(A)$.	
Mini masterclass structure (30 min)			
Activate (4 min): Roll a die. Ask for all the possible outcomes (the sample space). Model (12 min): Sample space {1,2,3,4,5,6}. $P(\text{rolling a 6}) = \frac{1}{6}$, so $P(\text{not a 6}) = 1 - \frac{1}{6} = \frac{5}{6}$. Complementary events add to 1. Guided practice (10 min): Students list sample spaces and find probabilities of complementary events for dice, spinners and bags. Check (4 min): Exit question: $P(\text{red}) = \frac{1}{4}$. What is $P(\text{not red})$?			
Key vocabulary		Differentiation	
sample space, outcome, probability, complementary event, not, total, 1, certain		Support: List the outcomes first; use 'everything minus the event' for the complement. Extension: Find complementary probabilities for two-step experiments.	
Assessment		Teaching notes (annotate here)	
Students find complementary probabilities — links to G3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Syllabus outcomes and content from Mathematics K-10 Syllabus (2022) copyright NSW Education Standards Authority (NESA) for and on behalf of the Crown in right of the State of New South Wales, 2024.

Unit 36 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Recognise the whole; compare halves and quarters of different-sized wholes.		
Video	Fractions — Understanding the Language Math with Mr. J	https://www.youtube.com/watch?v=38KfKBpwAv4
Video	Fractions — A Mini Review Course Math with Mr. J	https://www.youtube.com/watch?v=9UFyLHY8yPc
Worksheet (print)	Corbettmaths — Fractions (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-pdf.pdf
R2 -- Recall benchmark fraction-decimal-percentage equivalence.		
Video	Converting Between Fractions, Decimals, and Percents Math with Mr. J	https://www.youtube.com/watch?v=2vjbNZaFz3c
Video	Modeling Equivalent Fractions Math with Mr. J	https://www.youtube.com/watch?v=plognJLah-I
Worksheet (print)	Corbettmaths — Fractions, Decimals, Percentages (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf
R3 -- Find a half, quarter, fifth or tenth of a whole number.		
Video	How to Find Fractions of Amounts (Step-by-Step) Maths with Mrs B	https://www.youtube.com/watch?v=Md4Oeb_xDUk
Video	How to Find a Fraction of a Whole Number The Organic Chemistry Tutor	https://www.youtube.com/watch?v=ZPtxMHtXCM4
Worksheet (print)	Corbettmaths — Fractions of Amounts (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-of-amounts-pdf.pdf
R4 -- Subtract a unit fraction from a whole number (complement principle).		
Video	Subtracting a Fraction from a Whole Number Math with Mr. J	https://www.youtube.com/watch?v=xCcld_YyVhM
Video	Subtracting Fractions with Like Denominators Math with Mr. J	https://www.youtube.com/watch?v=QL4KiVVmrel
Worksheet (print)	Corbettmaths — Fractions (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-pdf.pdf
Y1 -- Build the whole quantity from a non-unit fractional part.		
Video	Fractions: Finding the Original Corbettmaths	https://www.youtube.com/watch?v=pdUg0UwXBHk
Video	Fractions of Numbers: Finding the Whole Partners in Prime	https://www.youtube.com/watch?v=x0ljSSMfZXw
Worksheet (print)	Corbettmaths — Fractions of Amounts (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-of-amounts-pdf.pdf
Y2 -- Compare fractions with related denominators using models.		
Video	Comparing Fractions How to Compare Fractions Math with Mr. J	https://www.youtube.com/watch?v=u_QTuWj107o
Video	Modeling Equivalent Fractions Math with Mr. J	https://www.youtube.com/watch?v=plognJLah-I
Worksheet (print)	Corbettmaths — Equivalent Fractions (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/equivalent-fractions-pdf.pdf
Y3 -- Order common fractions with related denominators.		
Video	Ordering Fractions Math with Mr. J	https://www.youtube.com/watch?v=i2PfHds88YE
Video	Comparing Fractions How to Compare Fractions Math with Mr. J	https://www.youtube.com/watch?v=u_QTuWj107o
Worksheet (print)	Corbettmaths — Ordering Fractions (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/ordering-fractions-pdf.pdf
Y4 -- Add and subtract fractions with related denominators using equivalence.		
Video	Subtracting Fractions with Common Denominators (Step by Step) Math with Mr. J	https://www.youtube.com/watch?v=VTCOHFJOAA8
Video	Adding & Subtracting Fractions with Unlike Denominators Math with Mr. J	https://www.youtube.com/watch?v=XsW8HJutlgM
Worksheet (print)	Corbettmaths — Adding Fractions: Same Denominators (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Fractions-Addition-1-pdf.pdf
Y5 -- Solve word problems adding or subtracting fractions with related denominators.		

Type	Resource	Link
Video	Adding and Subtracting Fractions Word Problems Math with Mr. J	https://www.youtube.com/watch?v=JgQSMwNOWCI
Video	Subtracting Fractions — Fraction Word Problems Math with Mr. J	https://www.youtube.com/watch?v=sxu_i1_SMqM
Worksheet (print)	Corbettmaths — Fractions (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-pdf.pdf
Y6 -- Assign probabilities using fractions and benchmark FDP.		
Video	An Intro to Probability Basic Probability Math with Mr. J	https://www.youtube.com/watch?v=6ldNYdA6no8
Video	An Intro to Finding Probability Math with Mr. J	https://www.youtube.com/watch?v=zE-Y_QCHOt8
Worksheet (print)	Corbettmaths — Probability (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf
G1 -- Add and subtract fractions with different denominators.		
Video	Adding & Subtracting Fractions with Unlike Denominators Math with Mr. J	https://www.youtube.com/watch?v=XsW8HJutlgM
Video	Adding Fractions with Unlike Denominators Math with Mr. J	https://www.youtube.com/watch?v=CoCmsyEQ_Xc
Worksheet (print)	Corbettmaths — Adding Fractions (Different Denominators) (PDF)	https://corbettmaths.com/wp-content/uploads/2024/01/Adding-Fractions.pdf
G2 -- Multiply a fraction by a whole number and by another fraction.		
Video	Multiplying Whole Numbers and Fractions Math with Mr. J	https://www.youtube.com/watch?v=2gFCEY9Hxas
Video	How to Multiply a Whole Number by a Fraction Math with Mr. J	https://www.youtube.com/watch?v=Z6liyRWnL4
Worksheet (print)	Corbettmaths — Multiplying Fractions (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/multiplying-fractions-pdf.pdf
G3 -- List the sample space and find probabilities of complementary events.		
Video	Calculating Probability (Practice Problems Included) Math with Mr. J	https://www.youtube.com/watch?v=m95FVOcCx44
Video	Complementary Events in Probability Math and Stats Help	https://www.youtube.com/watch?v=hJbj7UQIJNw
Worksheet (print)	Corbettmaths — Probability (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	Adding & Subtracting Fractions with Unlike Denominators Math with Mr. J	https://www.youtube.com/watch?v=XsW8HJutlgM
Video	Multiplying Whole Numbers and Fractions Math with Mr. J	https://www.youtube.com/watch?v=2gFCEY9Hxas
Video	Calculating Probability Math with Mr. J	https://www.youtube.com/watch?v=m95FVOcCx44
Video	Complementary Events in Probability Math and Stats Help	https://www.youtube.com/watch?v=hJbj7UQIJNw
Worksheet (print)	Corbettmaths — Adding Fractions (Different Denominators) (PDF)	https://corbettmaths.com/wp-content/uploads/2024/01/Adding-Fractions.pdf
Worksheet (print)	Corbettmaths — Multiplying Fractions (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/multiplying-fractions-pdf.pdf
Worksheet (print)	Corbettmaths — Probability (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf
Website	Maths is Fun — Adding Fractions (interactive)	https://www.mathsisfun.com/fractions_addition.html
Website	Maths is Fun — Probability (interactive)	https://www.mathsisfun.com/data/probability.html

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Fractions (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/fractions-pdf.pdf>
- ☐ Corbettmaths — Fractions, Decimals, Percentages (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf>
- ☐ Corbettmaths — Fractions of Amounts (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/fractions-of-amounts-pdf.pdf>
- ☐ Corbettmaths — Equivalent Fractions (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/equivalent-fractions-pdf.pdf>
- ☐ Corbettmaths — Ordering Fractions (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/ordering-fractions-pdf.pdf>
- ☐ Corbettmaths — Adding Fractions: Same Denominators (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/11/Fractions-Addition-1-pdf.pdf>

- ☐ Corbettmaths — Probability (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf>
- ☐ Corbettmaths — Adding Fractions (Different Denominators) (PDF) -- <https://corbettmaths.com/wp-content/uploads/2024/01/Adding-Fractions.pdf>
- ☐ Corbettmaths — Multiplying Fractions (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/multiplying-fractions-pdf.pdf>